

CHAPTER 7

Sheaves of Germs: The Topological Case

7.1 Localization over a Topological Space

The topological case refers to a localization scheme on a global partial order of physical events. These events are defined as taking place over a base category of reference contexts $O(X)$, consisting of open sets U of a topological measurement space X , the arrows between them being inclusions. In this case, the reference contexts of the operational environment employed for observation are all the open sets U of X , *partially ordered by inclusion*. Equivalently, the open set inclusions $U^\circ X$ are considered as varying base reference frames of open loci over which a global partial order of events fibers. We may use the suggestive term ‘local observer’ to refer to an event-registering device associated with a reference context U of the base category $O(X)$. Note that the meaning of ‘local’ here is understood with respect to the topology of X , such that a ‘local observer’ in a measurement situation taking place over a reference context U individuates events by means of local realvalued observables, being continuous maps $s : U \rightarrow \mathbb{R}$. Thus, the ‘local observers’ *do not* have a global ‘perception’ of continuous functions $f : X \rightarrow \mathbb{R}$, but rather always register events as localized over the associated reference contexts in terms of local observables. Of course, appropriate conditions are further needed for gluing their findings together; as will be explained presently, these conditions are the necessary and sufficient conditions for a topological sheaf-theoretic structure.

7.2 Uniform Localization of Observables

In order to clarify the functioning of a *topological localization process*, we will describe in detail the important case of localization of a commutative, unital $\mathbb{R}$ -algebra of observables of some system over a

base localizing category $O(X)$. Since observables are thought of as *global functions* on the R -coordinatized state-space of this system, the process of localization forces the replacement of the algebra of global observables A by an algebraic structure which will give us *all local and global functional information* together. All these functional elements should *interlock compatibly* together in an appropriate manner, which is characterized by the property that the combined structure respects the extension from local to global, as well as the restriction from global to local implied by the localization process. This structure is precisely formalized by the notion of a *sheaf of germs* of a commutative, unital R -algebra of observables, denoted by $\mathbf{A}$, which incorporates all compatible local and global information together.

Let us first introduce precisely the general notion of a sheaf on a topological space, and then examine its applicability in the current situation. For this purpose, we consider the category of open sets $O(X)$ in a topological space, partially ordered by inclusion. If $O(X)^{op}$ is the opposite category of $O(X)$, and $Sets$ denotes the category of sets, we define:

A *presheaf* of sets on $O(X)$ is a contravariant set-valued functor on $O(X)$, denoted by $\mathbf{P} : O(X)^{op} \rightarrow Sets$. For each base open set U of $O(X)$, $\mathbf{P}(U)$ is a set, and for each arrow $F : V \rightarrow U$, $\mathbf{P}(F) : \mathbf{P}(U) \rightarrow \mathbf{P}(V)$ is a set-function. If $\mathbf{P}$ is a presheaf on $O(X)$ and $p \in \mathbf{P}(U)$, the value $\mathbf{P}(F)(p)$ for an arrow $F : V \rightarrow U$ in $O(X)$ is called the *restriction* of p along F and is denoted by $\mathbf{P}(F)(p) := p \cdot F$. A presheaf $\mathbf{P}$ may be understood as *aright action* of $O(X)$ on a set. This set is partitioned into sorts parameterized by the objects of $O(X)$, and has the following property: If $F : V \rightarrow U$ is an inclusion arrow in $O(X)$ and p is an element of $\mathbf{P}$ of sort U , then $p \cdot F$ is specified as an element of $\mathbf{P}$ of sort V . Such an action $\mathbf{P}$ is referred to as an $O(X)$ *variable set*. A variable set of this form is entirely determined by its category of elements.

The *category of elements of a presheaf* $\mathbf{P}$, denoted by $j(\mathbf{P}, O(X))$, is described as follows: The objects of $j(\mathbf{P}, O(X))$ are all pairs (U, p) , with U in $O(X)$ and $p \in \mathbf{P}(U)$. The arrows of $j(\mathbf{P}, O(X))$, that is, $(U', p') \rightarrow (U, p)$, are those morphisms $Z : U' \rightarrow U$ in $O(X)$, such that $p' = \mathbf{P}(Z)(p) := p \cdot Z$. Notice that the arrows in $j(\mathbf{P}, O(X))$ are those morphisms $Z : U' \rightarrow U$ in the base category $O(X)$ that pull a chosen element $p \in \mathbf{P}(U)$ back into $p' \in \mathbf{P}(U')$.

The category of elements $j(\mathbf{P}, O(X))$ of a presheaf $\mathbf{P}$, together with the projection functor $\int_{\mathbf{P}} : j(\mathbf{P}, O(X)) \rightarrow O(X)$, is called the *split discrete fibration* induced by $\mathbf{P}$ ¹, where $O(X)$ is the *base category* of the fibration. We note that the fibers are categories in which the only arrows are identity arrows. If U is an open reference context of $O(X)$, the inverse image of U under $\int_{\mathbf{P}}$ is simply the set $\mathbf{P}(U)$, although its elements are written as pairs so as to form a *disjoint union*.

From a physical viewpoint, the purpose of introducing the notion of a presheaf $\mathbf{P}$ on $O(X)$ is the following: We identify an element of $\mathbf{P}$ of sort U , that is $p \in \mathbf{P}(U)$, with a *local observable*, which can be observed by means of a measurement procedure over the reference context U , being an open set of a topological space X . This identification forces the interrelations of local observables, over all reference contexts of the base category $O(X)$, to fulfill the requirements of a uniform and homologous fibred categorical structure. The latter are to be understood according to the following two requirements: First, the reference contexts used for observational purposes, together with their structural morphisms, should form a mathematical category. The notion of uniformity implies that for any two local observables, being amenable to a measurement procedure over the same open domain of measurement U , the structure of all reference contexts that relate to the first cannot be distinguished, in any possible way, from the structure of contexts relating to the second. According to this, all the localized observables within any particular reference context should be *uniformly equivalent* to each other. The split discrete fibration induced by $\mathbf{P}$, where $O(X)$ is the base category of the fibration, provides a well-defined notion of a *uniform homologous fibred structure* of local observables in the following two ways: [1] By the arrows specification defined in the category of elements of $\mathbf{P}$, any local observable p , determined over the reference context U , is homologically related with any other local observable p' over the reference context U' , and so on, by variation over all the reference contexts of the base category; [2] All the local observables p of $\mathbf{P}$, of the same sort U , viz., determined over the same reference context U , are uniformly equivalent to each other; this is because all the arrows in $\int(\mathbf{P}, O(X))$ are induced by lifting arrows from the base category $O(X)$, being formed by partially ordering the reference contexts. We conclude that the topological localization process is consistent with the physical requirement of uniformity.

7.3 Gluing Sections and Local-to-Global Compatibility

The next crucial step of the construction aims at the satisfaction of a second physical requirement: Since we have assumed the existence of reference contexts (open observational domains) locally, according to the operational requirements of a corresponding physical procedure of measurement, the information gathered about local observables in different measurement situations should be collated together by appropriate means. Mathematically, this requirement is implemented via the completion of the *presheaf* $\mathbf{P}$, or equivalently, sheafification of $\mathbf{P}$.²

A *sheaf* is characterized as a presheaf $\mathbf{P}$ that satisfies the following condition: If $U = \bigcup_a U_a$, U_a in $O(X)$, and elements $p_a \in \mathbf{P}(U_a)$, $a \in I$: index set, are such that for arbitrary $a, b \in I$, it holds:

$$p_a \mid U_{ab} = p_b \mid U_{ab}$$

where $U_{ab} := U_a \cap U_b$, and the symbol $\mid$ denotes the operation of *restriction* on the corresponding open reference context, then there exists a *unique* element $p \in \mathbf{P}(U)$, such that $p \mid U_a = p_a$ for each a in I .

Then an element of $\mathbf{P}(U)$ is called a *section* of the sheaf $\mathbf{P}$ over the open context U . The sheaf condition means that sections can be *glued together uniquely* over the reference contexts of the base category $\mathbf{O}(X)$. In particular, the sheaf-theoretic qualification of a uniform and homologous fibred structure of observables, as above, also makes the latter *coherent* in terms of *local-global compatibility* of the information content it carries, under the inverse operations of restriction and extension.

Thus, we form the following conclusion: The structure of a sheaf arises by imposing the following two requirements upon the uniform and homologous fibred structure of elements of the corresponding presheaf: [i] *Compatibility of observable information under restriction from the global to the local level*, and [ii] *Compatibility of observable information under extension from the local to the global level*. According to the first of the above requirements, a sheaf constitutes a *separated presheaf* of local observables over a global topological space, meaning that two observables are conceived as being identical globally if and only if they are identical locally. In turn, according to the second requirement, locally compatible observables can be glued together within some global observable, which is also uniquely defined because of the first requirement.

Furthermore, it is obvious that each set of sort U , $\mathbf{P}(U)$, can be endowed with the structure of an $\mathbf{R}$ -algebra under point-wise sum, product, and scalar multiplication, denoted correspondingly by $\mathbf{A}(U)$; in that case, the morphisms $\mathbf{A}(U) \rightarrow \mathbf{A}(V)$ stand for $\mathbf{R}$ -linear morphisms of $\mathbf{R}$ -algebras.

7.4 Contextuality: Germs and Stalks of

Observable Sheaves

Now we shall introduce the physically important notions of *stalks* and *germs* of a sheaf. For this purpose it is necessary to explain the construction of the *inductive limit* (categorical colimit) of sets (or $\mathbf{R}$ -algebras) of observables $\mathbf{A}(U)$, denoted by $\text{Colim}[\mathbf{A}(U)]$. Let us consider that x is a *point* of the topological measurement space X . Moreover, let K be a set consisting of open subsets of X , containing x , such that the following condition holds: For any two open reference contexts U, V , containing x , there exists an open reference context $W \in K$, *contained in the intersection* $U \cap V$. We may say that K constitutes a *basis* for the system of open reference contexts around x . We form the *disjoint union* of all $\mathbf{A}(U)$, denoted by:

$$\mathbf{D}(x) := \coprod_{U \in K} \mathbf{A}(U)$$

Then, we can define an *equivalence relation* in $\mathbf{D}(x)$ by requiring that $p : q$ for $p \in \mathbf{A}(U)$, $q \in \mathbf{A}(V)$, provided that they have the *same restriction to a smaller open set* contained in K . Then we define:

$$\text{Colim}_K[\mathbf{A}(U)] := \mathbf{D}(x) / :_K$$

Notice that if we denote the inclusion mapping of V into U by:

$$i_{V,U} : V \hookrightarrow U$$

and also the restriction morphism of sets from U to V by:

$$\rho_{U,V} : \mathbf{A}(U) \rightarrow \mathbf{A}(V)$$

we can introduce well-defined notions of addition and multiplication on the set $\text{Colim}_K[\mathbf{A}(U)]$ making it an $\mathbf{R}$ -algebra.

Now, if we consider that K and A are two bases for the system of open reference contexts around $x \in X$, we can show that there are canonical isomorphisms between $\text{Colim}_K[\mathbf{A}(U)]$ and $\text{Colim}_A[\mathbf{A}(U)]$. In particular, we may take all the open subsets of X containing x : Indeed, we consider first the case when K is arbitrary and A is the set of all open subsets containing x . Then $A \supset K$ induces a morphism:

$$\text{Colim}_K[\mathbf{A}(U)] \rightarrow \text{Colim}_A[\mathbf{A}(U)]$$

which is an isomorphism, because whenever V is an open subset containing x , there exists an open subset U in K contained in V . Since we can repeat that procedure for all bases of the system of open sets domains around $x \in X$, the initial claim follows immediately.

Then, the *stalk* of $\mathbf{A}$ at the point $x \in X$, denoted by $\mathbf{A}_x$, is defined as the *inductive limit* of sets (or $\mathbf{R}$ -algebras) of observables $\mathbf{A}(U)$:

$$\text{Colim}_K[\mathbf{A}(U)] := \coprod_{U \in K} \mathbf{A}(U) / :_K$$

where K is a basis for the system of open reference contexts around x , and $:_K$ denotes the equivalence relation of restriction within an open set in K . Note that this definition is *independent* of the chosen basis K . For an open reference context W containing the point x , we obtain an morphism of $\mathbf{A}(W)$ into the stalk at the point x :

$$i_{W,x} : \mathbf{A}(W) \rightarrow \mathbf{A}_x$$

For an element $p \in \mathbf{A}(W)$ its image:

$$i_{W,x}(p) := p_x = \text{germ}_x p$$

is called *the germ* of p at the point x . The fibred structure that corresponds to a sheaf of sets (or R-algebras) of observables $\mathbf{A}$ is a *topological bundle* defined by the *continuous* mapping $\varphi : A \rightarrow X$, where:

$$A = \coprod_{x \in X} \mathbf{A}_x$$

$$\varphi^{-1}(x) = \mathbf{A}_x = \text{Colim}_{\{x \in U\}} [\mathbf{A}(U)]$$

The mapping φ is locally a homeomorphism of topological spaces.³ The topology in A is defined as follows: for each $p \in \mathbf{A}(U)$, the set $\{p_x, x \in U\}$ is open, and moreover, an arbitrary open set is a union of sets of this form. In the physical state of affairs, we have identified an element of $\mathbf{A}$ of sort U that is a local section of $\mathbf{A}$, with a local observable p , which can be observed via a measurement procedure over the reference context U . Then the equivalence relation used in the definition of the stalk $\mathbf{A}_x$ at the point $x \in X$ is interpreted as follows: Two local observables $p \in \mathbf{A}(U)$ and $q \in \mathbf{A}(V)$ induce the same contextual information at x in X provided that they have the same restriction to a smaller open context contained in the basis K . Then the stalk $\mathbf{A}_x$ is the set containing all contextual information at x , that is, the set of all equivalence classes. Moreover, the image of a local observable $p \in \mathbf{A}(U)$ at the stalk $\mathbf{A}_x$, that is, the equivalence class of this local observable p , is precisely the germ of p at the point x .

Next, if we consider a local observable $p \in \mathbf{A}(U)$, it determines a function:

$$\dot{p} : x \mapsto \text{germ}_x p$$

whose domain is the open locus U and its codomain is the stalk $\mathbf{A}_x$ for each $x \in U$. Obviously, we may consider instead the disjoint union $A = \#_{x \in X} \mathbf{A}_x$ as the codomain of the function $\dot{p}$. In this sense, every local observable $p \in \mathbf{A}(U)$ gives rise to some partial function:

$$\dot{p} : U \rightarrow A$$

which is defined on the open reference context $U \subset X$. Hence, all local observables $p \in \mathbf{A}(U)$ admit a *functional representation*, established by means of the following correspondence:

$$\Delta(U): p \mapsto \dot{p}$$

Stated equivalently, each local observable $p \in \mathbf{A}(U)$ can be legitimately considered as a partial function:

$$\dot{p}: U \rightarrow A$$

defined over the open reference locus U , the value of which, at a point $x \in U$, viz., $\text{germ}_x p$, is the *contextual observable information induced at x* by the local observable p . Furthermore, such a partial function $\dot{p}: U \rightarrow A$ is identified with a *cross section of the topological bundle of germs*, defined by the continuous mapping $\varphi: A \rightarrow X$, such that:

$$\varphi^{-1}(x) = \mathbf{A}_x = \text{Colim}_{\{x \in U\}} [\mathbf{A}(U)]$$

Notice that the mapping φ is *locally* a homeomorphism of topological spaces, and thus the bundle is a topological *display bundle* (called equivalently an *étale bundle*.) Now, as a consequence of the *equivalence* between sheaves on a topological space X and display topological bundles over X , every *sheaf can be considered as a sheaf of cross sections of the equivalent display topological bundle*.⁴

7.5 Completion and Functionalization:

Display Bundles

It is also instructive to notice that the previous arguments can help us to understand the process of *completion* (or *sheafification*, or *germification*) of a presheaf.⁵ For this purpose, we realize that the notions of germ, stalk and display bundle make sense for a general presheaf. More precisely, the germ at a point stands for an equivalence class of elements of the presheaf corresponding to open reference contexts around that point, under the equivalence relation of having the same germ. The stalk over this point is the set of all germs at this point. The display bundle is the disjoint union of all stalks.

The first crucial observation is that by the definition of a topology on the display bundle, as described previously, it is legitimate to consider *continuous sections* of the display bundle. Stated equivalently, this procedure amounts to transforming the elements of the presheaf into partial continuous functions (continuous sections) valued into the display space. Hence, we manage to *functionalize* the initial presheaf, by defining a new presheaf, called the *presheaf of sections of the initial presheaf* as follows: *It is the presheaf, which associates to each open locus of the base topological space the set of continuous sections from that open locus into the display space*. Now, there is an obvious morphism from the initial presheaf to its presheaf of sections—a morphism that maps each element of the category

of elements of the initial presheaf to the continuous section, thus sending each point in an open reference context of the base space to the germ of this element at that point.

The second crucial observation is that *the associated functionalized presheaf* of sections of a presheaf is actually also *localized* (locally determined), viz., it is a sheaf, identified as the *sheaf of cross sections* of the corresponding topological display bundle. Thus, the latter sheaf is called the *sheaf associated with the initial presheaf*. Moreover, the process of completion of a presheaf into the sheaf of crosssections of the corresponding topological display bundle is *functorial*, meaning that there exists a functor sending each presheaf to its completion, viz., to its associated sheaf of sections, called the *sheafification functor*. As a corollary, we conclude that *a presheaf is a sheaf, viz., a complete presheaf, if and only if the morphism to its associated functional presheaf of sections is an isomorphism*. Thus, the process of *completion* of a presheaf is equivalent to the combined processes of *functionalization and localization* of its elements. Consequently, the associated sheaf of sections of a corresponding presheaf contains, by its construction, the totality of *local contextual information* compatible with that given in the initial presheaf via its restriction property. In this sense, the sheaf constitutes the completion of the presheaf.

7.6 Events: From Set-Theoretic to Sheaf-Theoretic Semantics

Now, let us consider a sheaf of R-algebras of local observables, identified as a sheaf of real-valued continuous cross-sections of the corresponding display bundle. Then, the set of germs of all these sections at a point, viz., the stalk at this point, is also an R-algebra. Most important, the stalk at this point is a *local* R-algebra, meaning that it has a *unique maximal ideal*. In turn, this maximal ideal consists of all germs vanishing at the considered point. The quotient of the stalk by this maximal ideal is isomorphic to the field of real numbers. Equivalently, this means that the morphism evaluating a germ of the stalk at a point to the real numbers (viz., providing a real value at the corresponding non-vanishing equivalence class of sections at the considered base point) is a surjective morphism of R-algebras having for a kernel the maximal ideal of the stalk at this point:

$$\mathbf{ev}_x : \mathbf{A}_x \rightarrow \mathbf{A}_x / \mu_x \cong \mathbb{R}$$

$$\mathit{germ}_x p \mapsto \mathbf{ev}_x(\mathit{germ}_x p) = p(x)$$

Thus, *the evaluation morphism of a germ* of the stalk at a point of the base space is an $\mathbb{R}$ -valued measurement of this observable germ, interpreted as an *observed event* of the corresponding system, and subsequently encoded by means of an $\mathbb{R}$ -state of its topological state-space.

Proceeding to the next stage of the development of this framework, the sheaf of germs of real-valued continuous functions on a topological space X is an object in the functor category of sheaves $\mathbf{Sh}(X)$ on varying reference contexts U , being open sets of X , partially ordered by inclusion. The morphisms in $\mathbf{Sh}(X)$ are all natural transformations between sheaves.⁶ It is instructive to notice that a sheaf makes sense only if the *base category of reference contexts* is specified, which is equivalent to the determination of a *topology* on the space X .

The functor category of sheaves $\mathbf{Sh}(X)$ provides an exemplary case of a construct known as *topos*.⁷ A topos can be conceived as a *local mathematical framework* corresponding to a generalized model of set theory, or as a *generalized algebraic space*, corresponding to a categorical universe of variable observable information sets over the multiplicity of the reference contexts of the base category. We recall that formally a topos is a category, which has a terminal object, pullbacks, exponentials, and a subobject classifier, which is understood as an object of generalized truth values. The particular significance of the sheaf of real-valued continuous functions on X is due to the following isomorphism: *The sheaf of germs of continuous real-valued functions on X is isomorphic to the object of Dedekind real numbers in the topos of sheaves $\mathbf{Sh}(X)$.*⁸ This isomorphism validates the physical intuition of considering a local observable as a *continuously variable real number* over its context of definition.

The transition in the semantics of the ‘physical event continuum’ from the topos of **Sets** to the topos of sheaves $\mathbf{Sh}(X)$ is an instance of the basic idea of *sheaf-theoretic localization of physical observability* referring to the interpretation of observed events. We initially notice that in the former case observables are identified with (continuous) functions determined completely by their values at points. In the latter case observables are identified with local continuous sections of the display space determined completely by their germs.

In order to analyze in more detail the transition in the semantics, we note that in the former case a continuous function from a base topological space X to the topological space $\mathbb{R}$ can be considered as a continuous section from X to the product space $X \times \mathbb{R}$. This product space is isomorphic set-theoretically to a space having a copy of $\mathbb{R}$ at each point, being the inverse image of the projection from $X \times \mathbb{R}$ to the base X . The value that is taken at a point is the value taken by the function. Thus, this type of modeling of an observable is only appropriate in capturing its *point properties*.

In contradistinction, in the latter case we strive to capture *local properties* of observables. This is because in the sheaf-theoretic local environment, we associate *not the value that a section takes at a*

point of the base space but its germ. In this sense, instead of the product total space $X \times \mathbb{R}$, we have the topological display space, such that the inverse image of each point of the base space is not a copy of $\mathbb{R}$, but the *stalk* at that point. This essentially means that the *transition of semantics* from the topos of sets **Sets** to the topos of sheaves $\mathbf{Sh}(X)$ amounts to a shift of focus from point-wise descriptions of observables to local descriptions of observables.

Obviously the topological display space is a much richer and larger space than the rigid space $X \times \mathbb{R}$, since the display space provides information about the *local behavior* of observables around each point of the base space in terms of germs, instead of merely point-wise behavior of observables in terms of their values in the real numbers. Thus, *observed events are not determined by the values of continuous functions (observables) at points of the base space, but by the evaluation morphisms of observable germs at those points*, according to our previous remarks.

We conclude that the meaning of the idea of sheaf-theoretic localization of physical observability referring to the interpretation of observed events, effectuated by the transition from the topos of sets **Sets** to the topos of sheaves $\mathbf{Sh}(X)$, amounts to a *topological relativization* with respect to the *local behavior* of physical observables as opposed to their point behavior.

Let us now further elaborate the sheaf-theoretic semantics of observed events from a process-theoretic standpoint. We argue that each event is a *novel actualized fact* and respects the selection constraints of the local context with respect to which it can be individuated. Thus events are essentially individuated and actualized with respect to local reference contexts U in $\mathcal{O}(X)$. It is crucial to notice that an event is not specified sheaf-theoretically by the evaluation of an observable, but rather by the evaluation of a corresponding observable germ; that is, an event is specified in a global sense by the *equivalence class of all compatible* local reference contexts conceived in all different local levels nested within each other with respect to which observable information can be glued.

Thus an observable germ is precisely the semantic information carrier of a *seed of extension* in the ontological formation of the continuum; it is, in other words, the germ of extensive connection from the local to the global via the topological gluing procedure. Notice also that the notion of a germ is *not* defined in abstraction from its local context, since its very definition makes sense only with respect to such a local context, where local sections agree. Moreover in the conception of a germ, a local context is *never* considered apart from the local information content which is contextualized. It is thus useful to think of germs as representing the *internal constitution of actual occasions* in their process of *concrecence*, since they integrate compatible information at higher and higher local levels (reference

contexts) by the formation of equivalence classes. In this way, it is via the concept of the germ that an actual occasion's objective global relevance can be analyzed—that is, via the semantic transition from the local to the global.

In Whiteheadian philosophy, an actual occasion is considered the starting point of its own process of concrescence, and this process is analyzable 'genetically.' In terms of the sheaf theoretic framework presented herein, the genetic division constitutive of an actual occasion is perfectly represented via the concept of the germ. Thus an actual occasion determines its germ of local observables or local sections. This, in turn, involves the determination of an appropriate local context with respect to which the germ is initially defined and eventually evaluated or actualized. The actualization or 'satisfaction' of the actual occasion's process of concrescence, in other words, can be understood in the context of quantum measurement as the evaluation / actualization of this germ as a quantum event.

7.7 Covering Sieves as Representable Functors

Before explaining the applicability of sheaf-theoretic localization of physical observability to the concept of a 'quantum event continuum,' it is important to first revisit and clarify the notions of sieves and covering sieves introduced in Chapter 6 as they pertain to the topological localization process discussed above. A U -sieve with respect to a reference context U in $O(X)$, is defined as a family S of $O(X)$ -morphisms with codomain U , such that if $V \rightarrow U$ belongs to S and $D \rightarrow V$ is any $O(X)$ -morphism, then the *nested composite inclusions (internal relations among local reference contexts)* $D \rightarrow V \rightarrow U$ belongs to S . Now, since all $O(X)$ -morphisms are inclusions, a U -sieve with respect to U is actually a *downwards closed* U -subset.

If we consider the contravariant representable functor of U in $O(X)$, viz., *the network of all internal relations that the reference context U has with all the other contexts in $O(X)$* , denoted by $\mathbf{y}[U] := \text{Hom}_{O(X)}(-, U)$, then it is easy to realize that a U -sieve is equivalent to a subfunctor $\mathbf{S} \# \mathbf{y}[U]$ in $\text{Sets}^{O(X)op}$.

In detail, given a U -sieve S , we define:

$$\mathbf{S}(V) = \{g / g: V \rightarrow U, g \in S\} \subseteq \mathbf{y}[U](V)$$

This definition yields a functor $\mathbf{S}$ in $\text{Sets}^{O(X)op}$, which is obviously a subfunctor of $\mathbf{y}[U]$. Conversely, given a subfunctor $\mathbf{S} \# \mathbf{y}[U]$ in $\text{Sets}_{O(X)op}$, the set:

$$S = \{g / g: V \rightarrow U, g \in \mathbf{S}(V)\}$$

for some reference context V in $O(X)$, is a U -sieve. Thus, we conclude:

$$\langle U\text{-sieve: } S \rangle = \langle \text{Subfunctor of } \mathbf{y}[U]: \mathbf{S}^{\circ} \rightarrow \mathbf{y}[U] \rangle$$

We notice that if S is a U -sieve and $h: V \rightarrow U$ is any inclusion to the reference context U , then:

$$h^*(S) = \{f \mid \text{cod}(f) = V, (h \circ f) \in S\}$$

is a V -sieve, called the *pullback* of S along h . Consequently, we may define a presheaf functor $\mathbf{K}$ in $\text{Sets}_{O(X)_{op}}$, such that its action on reference contexts U in $O(X)$, is given by:

$$\mathbf{K}(U) = \{S \mid S : U \text{ - sieve}\}$$

and on arrows $h: V \rightarrow U$, by $h^*(\#): \mathbf{K}(U) \rightarrow \mathbf{K}(V)$, given by:

$$h^*(S) = \{f \mid \text{cod}(f) = V, (h \circ f) \in S\}$$

We notice that for a reference context U in $O(X)$, the set of all inclusion arrows into U , called the *maximal sieve* on U , and denoted by $t(U) := t_U$, is a U -sieve. Moreover, if we consider the partially ordered set of open subsets of a topological space X , viewed as the base category of open reference contexts, $O(X)$ as above, then a U -sieve S (where U is an open reference context) is a covering U -sieve if and only if U is contained in the union of open sets in S . This fulfills the requirements of covering sieves, and consequently, defines a *Grothendieck topology* J on $O(X)$. A Grothendieck topology J exists as a presheaf functor $\mathbf{K}_X$ in $\text{Sets}^{O(X)_{op}}$, such that by acting on reference contexts U in $O(X)$, J gives the set of all covering U -sieves, denoted by $\mathbf{K}_X(U)$; whereas by acting on figures $h: V \rightarrow U$, it gives a morphism $h^*(\#): \mathbf{K}_X(U) \rightarrow \mathbf{K}_X(V)$, expressed as:

$$h^*(S) = \{f \mid \text{cod}(f) = V, (h \circ f) \in S\}, \text{ for } S \in \mathbf{K}_X(U).$$

Notes

1. S. Mac Lane and I. Moerdijk, *Sheaves in Geometry and Logic* (New York: Springer-Verlag, 1992).
2. Mac Lane and Moerdijk, *Sheaves in Geometry and Logic*. See also: A. Mallios, *Geometry of Vector Sheaves: An Axiomatic Approach to Differential Geometry*, Vol. 1 (Kluwer Academic Publishers, Dordrecht, 1998).
3. To each a in A there exists an open set V , with $a \in V \subseteq A$, such that $\varphi(V)$ is open in X and $\varphi|_V$ is a homeomorphism (continuous function that has a continuous inverse) $V \rightarrow \varphi(V)$.
4. Mac Lane and Moerdijk, *Sheaves in Geometry and Logic*. See also: Mallios, *Geometry of Vector Sheaves*, and A. Mallios, "On Localizing Topological Algebras," *Contemporary Mathematics* 341, no. 79 (2004).
5. Mallios, "On Localizing Topological Algebras."

6. Mac Lane and Moerdijk, *Sheaves in Geometry and Logic*.
7. S. Awodey, *Category Theory*, Oxford Logic Guides (Oxford: Oxford University Press, 2006). See also: Mac Lane and Moerdijk, *Sheaves in Geometry and Logic*.
8. Mac Lane and Moerdijk, *Sheaves in Geometry and Logic*.